
Historical Performance Test Executions

Scenario baselines, failed executions, regression evidence, and historical comparison records

Corpus ID	PTA-HIST-1500-V2
Version	2.0 - Expanded RAG Edition
Source dataset	performance_test_dataset_1500_unique_names.csv
Rows represented	1,500
Primary consumer	Analysis Agent inside the Performance Test Agent

Document purpose

- Provide historical evidence for the Analysis Agent when it compares a current run with similar executions.
- Represent all five scenarios and all five bottleneck categories found in the uploaded 1,500-row dataset.
- Supply chunk-friendly incident narratives containing symptoms, evidence, root cause, and remediation.
- Support queries such as: find similar failures, compare scenario baselines, and identify repeated bottlenecks.

Data handling note: This document uses aggregate metrics and selected Job IDs. Customer names and account values are intentionally excluded because they do not improve performance root-cause analysis. All content is synthetic and intended for RAG development, demonstrations, and testing.

Recommended RAG metadata

Field	Example value
document_type	Historical Performance Test Executions
domain	banking-performance-testing
agent	analysis-agent
confidentiality	dummy-synthetic
chunk_strategy	section-aware, 500-900 tokens, 100-token overlap
retrieval_keys	scenario, API, status, symptom, bottleneck, recommendation, SLA

1. Corpus overview and aggregate baseline

A numerical map of the dataset before incident-level retrieval.

The uploaded dataset contains **1,500 execution records**. It includes **1,203 SLA passes** and **297 SLA failures**. The overall failure rate is **19.8%**. All successful records return HTTP 200, while failed records use HTTP 500, 502, 503, or 504.

Metric	Minimum	Median	P90	P95	Maximum	Mean
Response time (ms)	121	667	2778	3464	4178	1006.8
CPU (%)	35	65	93	96	99	65.9
Memory (%)	38	64	88	92	96	64.6
DB connections	60	172	417	466	520	202.2

Interpretation for retrieval

- A response time near 667 ms is typical for the full mixed dataset, but healthy and failed populations must be compared separately.
- The 90th percentile response time is about 2,778 ms, marking the region where severe failures dominate retrieval results.
- CPU above 90%, memory above 88%, or database connections above 417 belong to the upper decile and should be treated as strong symptoms.
- A retrieval query should include both the current scenario and the strongest abnormal metric, rather than searching only for generic failure text.

Scenario-level baseline

Scenario	Rows	SLA fail	Fail %	Avg RT	P95 RT	Avg CPU	Avg Mem	Avg DB
Fund Transfer	307	61	19.9%	1028	3617	64.8	64.1	203.2
IMPS Transfer	320	66	20.6%	1030	3529	66.5	64.8	202.7
NEFT Transfer	276	56	20.3%	989	3210	66.6	65.6	204.0
RTGS Transfer	292	62	21.2%	1024	3493	66.2	65.3	206.4
UPI Transfer	305	52	17.0%	961	3407	65.5	63.2	194.9

Baseline rule: compare a current execution first against its own scenario distribution, then against the global baseline. This avoids falsely labeling a naturally slower scenario as a regression.

2. UPI Transfer historical cases

Representative UPI failures selected from the dataset with varied bottleneck labels.

UPI-HIST-01

Field	Value	Field	Value
Job ID	JOB-20260801-0601	Scenario	UPI Transfer
API	UPI Transfer API	HTTP	503
Response	4,060 ms	SLA	FAIL
CPU	86%	Memory	81%
DB connections	411	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	very-high-value

Analysis narrative: This UPI execution failed with HTTP 503. The response time of 4,060 ms is far above the dataset median. CPU was 86% and memory was 81%, while the database used 411 connections. The recorded bottleneck is Database Pool. The recommended first action is Increase DB Pool Size. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

UPI-HIST-02

Field	Value	Field	Value
Job ID	JOB-20260801-1375	Scenario	UPI Transfer
API	UPI Transfer API	HTTP	504
Response	4,142 ms	SLA	FAIL
CPU	92%	Memory	83%
DB connections	505	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	high-value

Analysis narrative: This UPI execution failed with HTTP 504. The response time of 4,142 ms is far above the dataset median. CPU was 92% and memory was 83%, while the database used 505 connections. The recorded bottleneck is High CPU. The recommended first action is Scale Application Horizontally. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

UPI-HIST-03

Field	Value	Field	Value
Job ID	JOB-20260801-1115	Scenario	UPI Transfer
API	UPI Transfer API	HTTP	500
Response	4,064 ms	SLA	FAIL
CPU	96%	Memory	83%
DB connections	327	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	high-value

Analysis narrative: This UPI execution failed with HTTP 500. The response time of 4,064 ms is far above the dataset median. CPU was 96% and memory was 83%, while the database used 327 connections. The recorded bottleneck is Memory Pressure. The recommended first action is Increase JVM Heap Memory. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

UPI-HIST-04

Field	Value	Field	Value
Job ID	JOB-20260801-1417	Scenario	UPI Transfer

Field	Value	Field	Value
API	UPI Transfer API	HTTP	500
Response	4,063 ms	SLA	FAIL
CPU	87%	Memory	87%
DB connections	375	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	very-high-value

Analysis narrative: This UPI execution failed with HTTP 500. The response time of 4,063 ms is far above the dataset median. CPU was 87% and memory was 87%, while the database used 375 connections. The recorded bottleneck is Thread Contention. The recommended first action is Tune Thread Pool. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

UPI-HIST-05

Field	Value	Field	Value
Job ID	JOB-20260801-0795	Scenario	UPI Transfer
API	UPI Transfer API	HTTP	502
Response	3,503 ms	SLA	FAIL
CPU	92%	Memory	82%
DB connections	458	Bottleneck	Network Latency
Recommendation	Investigate Network Bottleneck	Amount band	very-high-value

Analysis narrative: This UPI execution failed with HTTP 502. The response time of 3,503 ms is far above the dataset median. CPU was 92% and memory was 82%, while the database used 458 connections. The recorded bottleneck is Network Latency. The recommended first action is Investigate Network Bottleneck. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

UPI-HIST-06

Field	Value	Field	Value
Job ID	JOB-20260801-0750	Scenario	UPI Transfer
API	UPI Transfer API	HTTP	504
Response	3,975 ms	SLA	FAIL
CPU	92%	Memory	85%
DB connections	436	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	high-value

Analysis narrative: This UPI execution failed with HTTP 504. The response time of 3,975 ms is far above the dataset median. CPU was 92% and memory was 85%, while the database used 436 connections. The recorded bottleneck is Thread Contention. The recommended first action is Tune Thread Pool. For a new UPI failure, retrieve this record when at least two metric bands and the HTTP status are similar.

3. Fund Transfer historical cases

Historical evidence for generic transfer requests and Transfer API behavior.

FUND-HIST-01

Field	Value	Field	Value
Job ID	JOB-20260801-0778	Scenario	Fund Transfer
API	Transfer API	HTTP	500
Response	3,933 ms	SLA	FAIL
CPU	96%	Memory	86%
DB connections	473	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	very-high-value

Analysis narrative: This Fund Transfer record indicates Database Pool under a 500 response. The execution reached 3,933 ms with CPU 96%, memory 86%, and 473 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Increase DB Pool Size.

FUND-HIST-02

Field	Value	Field	Value
Job ID	JOB-20260801-0799	Scenario	Fund Transfer
API	Transfer API	HTTP	504
Response	4,178 ms	SLA	FAIL
CPU	86%	Memory	95%
DB connections	478	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	low-value

Analysis narrative: This Fund Transfer record indicates High CPU under a 504 response. The execution reached 4,178 ms with CPU 86%, memory 95%, and 478 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Scale Application Horizontally.

FUND-HIST-03

Field	Value	Field	Value
Job ID	JOB-20260801-0203	Scenario	Fund Transfer
API	Transfer API	HTTP	500
Response	3,945 ms	SLA	FAIL
CPU	95%	Memory	90%
DB connections	418	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	low-value

Analysis narrative: This Fund Transfer record indicates Memory Pressure under a 500 response. The execution reached 3,945 ms with CPU 95%, memory 90%, and 418 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Increase JVM Heap Memory.

FUND-HIST-04

Field	Value	Field	Value
Job ID	JOB-20260801-0906	Scenario	Fund Transfer
API	Transfer API	HTTP	504
Response	3,918 ms	SLA	FAIL

Field	Value	Field	Value
CPU	87%	Memory	85%
DB connections	336	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	standard-value

Analysis narrative: This Fund Transfer record indicates Thread Contention under a 504 response. The execution reached 3,918 ms with CPU 87%, memory 85%, and 336 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Tune Thread Pool.

FUND-HIST-05

Field	Value	Field	Value
Job ID	JOB-20260801-0470	Scenario	Fund Transfer
API	Transfer API	HTTP	500
Response	4,067 ms	SLA	FAIL
CPU	87%	Memory	86%
DB connections	491	Bottleneck	Network Latency
Recommendation	Investigate Network Bottleneck	Amount band	very-high-value

Analysis narrative: This Fund Transfer record indicates Network Latency under a 500 response. The execution reached 4,067 ms with CPU 87%, memory 86%, and 491 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Investigate Network Bottleneck.

FUND-HIST-06

Field	Value	Field	Value
Job ID	JOB-20260801-0817	Scenario	Fund Transfer
API	Transfer API	HTTP	504
Response	3,927 ms	SLA	FAIL
CPU	95%	Memory	90%
DB connections	427	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	standard-value

Analysis narrative: This Fund Transfer record indicates High CPU under a 504 response. The execution reached 3,927 ms with CPU 95%, memory 90%, and 427 database connections. The event is useful as a historical neighbor when the live Transfer API shows a matching resource pattern. The stored remediation is: Scale Application Horizontally.

4. IMPS Transfer historical cases

Fast-payment workflow failures and their probable infrastructure signatures.

IMPS-HIST-01

Field	Value	Field	Value
Job ID	JOB-20260801-1086	Scenario	IMPS Transfer
API	IMPS API	HTTP	504
Response	4,094 ms	SLA	FAIL
CPU	89%	Memory	80%
DB connections	388	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	high-value

Analysis narrative: The IMPS request failed its SLA at 4,094 ms. HTTP 504, CPU 89%, memory 80%, and DB connections 388 form the evidence vector. The labeled cause is Database Pool, mapped to the action Increase DB Pool Size. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

IMPS-HIST-02

Field	Value	Field	Value
Job ID	JOB-20260801-1066	Scenario	IMPS Transfer
API	IMPS API	HTTP	502
Response	3,915 ms	SLA	FAIL
CPU	92%	Memory	81%
DB connections	398	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	standard-value

Analysis narrative: The IMPS request failed its SLA at 3,915 ms. HTTP 502, CPU 92%, memory 81%, and DB connections 398 form the evidence vector. The labeled cause is High CPU, mapped to the action Scale Application Horizontally. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

IMPS-HIST-03

Field	Value	Field	Value
Job ID	JOB-20260801-0017	Scenario	IMPS Transfer
API	IMPS API	HTTP	502
Response	4,031 ms	SLA	FAIL
CPU	93%	Memory	91%
DB connections	342	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	standard-value

Analysis narrative: The IMPS request failed its SLA at 4,031 ms. HTTP 502, CPU 93%, memory 91%, and DB connections 342 form the evidence vector. The labeled cause is Memory Pressure, mapped to the action Increase JVM Heap Memory. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

IMPS-HIST-04

Field	Value	Field	Value
Job ID	JOB-20260801-1090	Scenario	IMPS Transfer
API	IMPS API	HTTP	503
Response	4,174 ms	SLA	FAIL

Field	Value	Field	Value
CPU	88%	Memory	80%
DB connections	335	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	high-value

Analysis narrative: The IMPS request failed its SLA at 4,174 ms. HTTP 503, CPU 88%, memory 80%, and DB connections 335 form the evidence vector. The labeled cause is Thread Contention, mapped to the action Tune Thread Pool. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

IMPS-HIST-05

Field	Value	Field	Value
Job ID	JOB-20260801-1170	Scenario	IMPS Transfer
API	IMPS API	HTTP	504
Response	4,117 ms	SLA	FAIL
CPU	90%	Memory	82%
DB connections	363	Bottleneck	Network Latency
Recommendation	Investigate Network Bottleneck	Amount band	low-value

Analysis narrative: The IMPS request failed its SLA at 4,117 ms. HTTP 504, CPU 90%, memory 82%, and DB connections 363 form the evidence vector. The labeled cause is Network Latency, mapped to the action Investigate Network Bottleneck. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

IMPS-HIST-06

Field	Value	Field	Value
Job ID	JOB-20260801-0053	Scenario	IMPS Transfer
API	IMPS API	HTTP	500
Response	4,088 ms	SLA	FAIL
CPU	94%	Memory	87%
DB connections	503	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	low-value

Analysis narrative: The IMPS request failed its SLA at 4,088 ms. HTTP 500, CPU 94%, memory 87%, and DB connections 503 form the evidence vector. The labeled cause is Thread Contention, mapped to the action Tune Thread Pool. During retrieval, preserve the distinction between front-end network symptoms and back-end saturation symptoms.

5. NEFT Transfer historical cases

Batch-oriented transfer examples for comparison and regression diagnosis.

NEFT-HIST-01

Field	Value	Field	Value
Job ID	JOB-20260801-1265	Scenario	NEFT Transfer
API	NEFT API	HTTP	503
Response	3,925 ms	SLA	FAIL
CPU	89%	Memory	83%
DB connections	466	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	standard-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 3,925 ms and HTTP status is 503. Resource evidence includes CPU 89%, memory 83%, and 466 DB connections. The dataset assigns Database Pool and recommends Increase DB Pool Size. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

NEFT-HIST-02

Field	Value	Field	Value
Job ID	JOB-20260801-1097	Scenario	NEFT Transfer
API	NEFT API	HTTP	500
Response	3,993 ms	SLA	FAIL
CPU	93%	Memory	80%
DB connections	322	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	very-high-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 3,993 ms and HTTP status is 500. Resource evidence includes CPU 93%, memory 80%, and 322 DB connections. The dataset assigns High CPU and recommends Scale Application Horizontally. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

NEFT-HIST-03

Field	Value	Field	Value
Job ID	JOB-20260801-0059	Scenario	NEFT Transfer
API	NEFT API	HTTP	503
Response	4,132 ms	SLA	FAIL
CPU	88%	Memory	94%
DB connections	498	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	high-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 4,132 ms and HTTP status is 503. Resource evidence includes CPU 88%, memory 94%, and 498 DB connections. The dataset assigns Memory Pressure and recommends Increase JVM Heap Memory. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

NEFT-HIST-04

Field	Value	Field	Value
Job ID	JOB-20260801-0905	Scenario	NEFT Transfer
API	NEFT API	HTTP	500

Field	Value	Field	Value
Response	3,123 ms	SLA	FAIL
CPU	93%	Memory	96%
DB connections	488	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	very-high-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 3,123 ms and HTTP status is 500. Resource evidence includes CPU 93%, memory 96%, and 488 DB connections. The dataset assigns Thread Contention and recommends Tune Thread Pool. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

NEFT-HIST-05

Field	Value	Field	Value
Job ID	JOB-20260801-0468	Scenario	NEFT Transfer
API	NEFT API	HTTP	503
Response	3,606 ms	SLA	FAIL
CPU	98%	Memory	96%
DB connections	434	Bottleneck	Network Latency
Recommendation	Investigate Network Bottleneck	Amount band	high-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 3,606 ms and HTTP status is 503. Resource evidence includes CPU 98%, memory 96%, and 434 DB connections. The dataset assigns Network Latency and recommends Investigate Network Bottleneck. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

NEFT-HIST-06

Field	Value	Field	Value
Job ID	JOB-20260801-0239	Scenario	NEFT Transfer
API	NEFT API	HTTP	504
Response	4,037 ms	SLA	FAIL
CPU	86%	Memory	91%
DB connections	322	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	high-value

Analysis narrative: This NEFT execution is a failed historical sample. Response time is 4,037 ms and HTTP status is 504. Resource evidence includes CPU 86%, memory 91%, and 322 DB connections. The dataset assigns Memory Pressure and recommends Increase JVM Heap Memory. A future Analysis Agent should cite this job only when the scenario, symptoms, and temporal pattern are sufficiently similar.

6. RTGS Transfer historical cases

High-value transfer failures with cross-resource diagnostic context.

RTGS-HIST-01

Field	Value	Field	Value
Job ID	JOB-20260801-0414	Scenario	RTGS Transfer
API	RTGS API	HTTP	503
Response	4,139 ms	SLA	FAIL
CPU	89%	Memory	94%
DB connections	395	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	standard-value

Analysis narrative: The RTGS execution recorded 4,139 ms and HTTP 503. CPU was 89%, memory was 94%, and database connections reached 395. The historical label is Database Pool. The proposed fix is Increase DB Pool Size. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

RTGS-HIST-02

Field	Value	Field	Value
Job ID	JOB-20260801-0482	Scenario	RTGS Transfer
API	RTGS API	HTTP	502
Response	4,026 ms	SLA	FAIL
CPU	99%	Memory	88%
DB connections	369	Bottleneck	High CPU
Recommendation	Scale Application Horizontally	Amount band	low-value

Analysis narrative: The RTGS execution recorded 4,026 ms and HTTP 502. CPU was 99%, memory was 88%, and database connections reached 369. The historical label is High CPU. The proposed fix is Scale Application Horizontally. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

RTGS-HIST-03

Field	Value	Field	Value
Job ID	JOB-20260801-0038	Scenario	RTGS Transfer
API	RTGS API	HTTP	502
Response	3,946 ms	SLA	FAIL
CPU	92%	Memory	84%
DB connections	361	Bottleneck	Memory Pressure
Recommendation	Increase JVM Heap Memory	Amount band	low-value

Analysis narrative: The RTGS execution recorded 3,946 ms and HTTP 502. CPU was 92%, memory was 84%, and database connections reached 361. The historical label is Memory Pressure. The proposed fix is Increase JVM Heap Memory. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

RTGS-HIST-04

Field	Value	Field	Value
Job ID	JOB-20260801-0299	Scenario	RTGS Transfer
API	RTGS API	HTTP	500
Response	3,924 ms	SLA	FAIL

Field	Value	Field	Value
CPU	95%	Memory	81%
DB connections	431	Bottleneck	Thread Contention
Recommendation	Tune Thread Pool	Amount band	very-high-value

Analysis narrative: The RTGS execution recorded 3,924 ms and HTTP 500. CPU was 95%, memory was 81%, and database connections reached 431. The historical label is Thread Contention. The proposed fix is Tune Thread Pool. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

RTGS-HIST-05

Field	Value	Field	Value
Job ID	JOB-20260801-0579	Scenario	RTGS Transfer
API	RTGS API	HTTP	504
Response	4,084 ms	SLA	FAIL
CPU	97%	Memory	93%
DB connections	454	Bottleneck	Network Latency
Recommendation	Investigate Network Bottleneck	Amount band	very-high-value

Analysis narrative: The RTGS execution recorded 4,084 ms and HTTP 504. CPU was 97%, memory was 93%, and database connections reached 454. The historical label is Network Latency. The proposed fix is Investigate Network Bottleneck. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

RTGS-HIST-06

Field	Value	Field	Value
Job ID	JOB-20260801-0861	Scenario	RTGS Transfer
API	RTGS API	HTTP	504
Response	4,029 ms	SLA	FAIL
CPU	99%	Memory	87%
DB connections	434	Bottleneck	Database Pool
Recommendation	Increase DB Pool Size	Amount band	standard-value

Analysis narrative: The RTGS execution recorded 4,029 ms and HTTP 504. CPU was 99%, memory was 87%, and database connections reached 434. The historical label is Database Pool. The proposed fix is Increase DB Pool Size. RTGS retrieval should emphasize reliability and timeout signatures, especially when HTTP 504 co-occurs with network or thread symptoms.

7. Cross-scenario regression patterns

Patterns that repeat across APIs and should be stored as reusable historical lessons.

Pattern ID	Historical pattern	Evidence signature	Likely interpretation	Retrieval tags
REG-001	Latency cliff after a healthy region	Response > 2,500 ms, HTTP 5xx, CPU and memory rising	Capacity boundary or cascading back-pressure	latency-cliff, saturation, 5xx
REG-002	Database pressure without maximum CPU	DB connections > 400, CPU may be moderate, response > 2,000 ms	Connection wait or slow query accumulation	database-pool, connection-wait
REG-003	CPU-dominant failure	CPU >= 90%, memory not equally elevated	Compute saturation, hot code path, serialization, encryption, or loop	high-cpu, scale-out, profiling
REG-004	Memory-dominant failure	Memory >= 88%, long response, possible 502/503	Heap pressure, GC pause, cache growth, large payload retention	memory-pressure, gc, heap
REG-005	Timeout-dominant failure	HTTP 504, high response, resource signal may be mixed	Gateway timeout, downstream latency, network delay, or blocked worker	timeout, network, dependency
REG-006	Thread contention	HTTP 500/504, high latency, CPU variable	Worker pool exhausted, lock contention, or queue growth	thread-pool, lock, queue
REG-007	False similarity risk	Same HTTP code but different dominant resource	Do not infer root cause from status code alone	disambiguation, multi-signal

Historical comparison algorithm

- Filter by scenario or API when those fields are known.
- Compute metric bands for response time, CPU, memory, and database connections.
- Boost records with matching HTTP class and matching bottleneck vocabulary.
- Down-rank records that match only the status code but contradict the dominant resource signal.
- Return at least one similar case and one counterexample so the model can explain uncertainty.

8. Retrieval examples and expected Analysis Agent use

Example questions, useful chunks, and grounded answer patterns.

Example 1: Why did the current UPI test fail with CPU 94%, memory 70%, DB 180, HTTP 500?

Retrieval plan: Retrieve UPI + High CPU cases, CPU upper-decile records, and HTTP 500 examples. Avoid database-pool incidents because DB connections are not elevated.

Grounded response: Probable compute saturation. Recommend profiling hot methods and scaling application instances. Confidence increases if TPS plateaus while CPU remains above 90%.

Example 2: Current Fund Transfer has 3,200 ms, CPU 73%, memory 68%, DB 470, HTTP 504. What is the root cause?

Retrieval plan: Retrieve Fund Transfer + Database Pool and timeout patterns. DB connections are the dominant abnormal signal.

Grounded response: Probable database connection pool exhaustion or long-running queries causing queue wait. Check pool wait time, active versus idle connections, and slow SQL before increasing pool size.

Example 3: Response time increased but all resource values are normal. What should be checked?

Retrieval plan: Retrieve Network Latency and dependency timeout records, especially HTTP 502/504 examples.

Grounded response: Investigate network path, gateway, DNS, TLS negotiation, and downstream dependency latency. Do not recommend CPU or heap changes without evidence.

Example 4: How does this RTGS run compare with historical baseline?

Retrieval plan: Retrieve RTGS baseline table and nearest historical RTGS records.

Grounded response: Report delta from RTGS average and P95, SLA result, nearest incidents, and whether the observed bottleneck repeats a known pattern.

Suggested chunk metadata

Each incident chunk should retain: document_id, section_id, job_id, scenario, API, HTTP status, SLA, bottleneck, recommendation, metric bands, source file, and version. This allows hybrid retrieval using both embeddings and exact metadata filters.

9. Historical corpus maintenance and quality rules

How to keep the historical knowledge useful as the dataset grows.

- Add a new historical incident only after the test result and root-cause label have been reviewed.
- Keep raw measured values separate from human conclusions. The Analysis Agent should be able to distinguish evidence from interpretation.
- Do not overwrite old incidents after a configuration change. Create a new version so regressions and improvements remain traceable.
- Store the fix outcome: resolved, partially resolved, ineffective, or not validated.
- Record environment metadata such as application version, server count, database version, pool size, and test profile when available.
- Mask or omit customer names, account identifiers, tokens, secrets, and payload values before ingestion.
- Evaluate retrieval monthly using known questions and expected supporting incident IDs.

Example historical record schema

```
{
  "record_id": "HIST-FUND-042",
  "scenario": "Fund Transfer",
  "metrics": {"response_ms": 3210, "cpu_pct": 73, "memory_pct": 68, "db_connections": 470},
  "http_status": 504,
  "root_cause": "Database Pool",
  "recommendation": "Inspect pool wait time and slow SQL; increase pool only after capacity validation",
  "resolution_status": "validated",
  "source": "performance_test_dataset_1500_unique_names.csv"
}
```

End of historical execution corpus. This document is intended to be ingested alongside the structured CSV, not used as a replacement for it.
